

Notes: Harrison, Ross, and Swarthout (JPE, 2026)

Gender, Confidence, and the Mismeasure of Intelligence, Competitiveness, and Literacy

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1 Big picture

- **Question.** Do standard tests mismeasure intelligence because they ignore the *confidence* with which people hold answers, and does the same measurement error explain famous conclusions about women being less competitive or less financially literate?
- **Setting.** The paper studies laboratory experiments built around Raven Advanced Progressive Matrices (RAPM) questions. The core comparison is between traditional point-answer scoring and an incentivized system that lets subjects report full belief distributions over the eight possible answers to each Raven question.
- **Core challenge.** A forced multiple-choice answer confounds at least three things: actual knowledge, awareness of uncertainty, and willingness to express that uncertainty. On top of that, once answers have monetary consequences, risk preferences matter for how subjects should report beliefs.
- **Main conceptual contribution.** The paper imports proper scoring rules and welfare analysis into the measurement of intelligence. It argues that the right object is not just whether the modal answer is correct, but how much confidence the subject appropriately places on the correct answer.
- **Main empirical design.** Subjects are randomly assigned to five main Raven treatments: baseline pen-and-paper, One Token Progressive, 80 Tokens Progressive, One Token Scrambled, and 80 Tokens Scrambled. The one-token treatments force a modal-belief response; the 80-token treatments let subjects express confidence across alternatives.
- **Main performance metrics.** The paper distinguishes *accuracy* from *earnings efficiency*. Accuracy is a hit-rate style measure. Efficiency is realized earnings relative to the maximum possible under the scoring rule. This distinction is central.
- **Main baseline result.** Providing salient financial incentives raises Raven accuracy sharply: compared with the traditional baseline, the One Token Progressive treatment improves accuracy by 14.2 percentage points.
- **Main confidence result.** Allowing subjects to report confidence does *not* significantly increase raw accuracy relative to one-token reporting, but it does increase earnings efficiency by 17.5 percentage points. So the unconstrained response dominates once the correct performance metric is used.
- **Main gender result.** Standard modal-answer measurement makes women look weaker on Raven accuracy. Once confidence is elicited and rewarded, women do better than men on efficiency; Black subjects also improve markedly. The same qualitative result survives when the progressive ordering of Raven difficulty is scrambled.
- **Main scaffold result.** The elicitation interface itself acts as a *cognitive scaffold*. The paper's bold claim is not merely that the interface reveals preexisting intelligence more accurately, but that it can make subjects cognitively more effective on average.
- **Main competitiveness result.** Reinterpreting tournament-choice experiments as risky-lottery choices changes the gender conclusion. Once subjective win probabilities and risk preferences are used to compute certainty equivalents, women do not appear to “shy away” from competition too much; men more often choose tournaments when they should not.

- **Main literacy result.** Using an incentivized inflation-literacy question, the paper argues that women are not simply “less literate.” Women often appear to express uncertainty more appropriately. The normatively worrisome type is not a diffuse and cautious respondent but a confidently wrong one.
- **Main takeaway.** The paper’s deepest message is methodological: intelligence, competitiveness, and literacy should be measured with explicit incentives, explicit confidence reporting, and welfare-relevant evaluation. “Appropriate confidence” matters more than “more confidence.”

2 Data and measurement (key objects)

2.1 Why the Raven task is useful

- The paper uses the Raven Advanced Progressive Matrices test, especially Set II, which contains 36 problems with eight candidate answers each.
- Raven problems are attractive because they move gradually from easy to hard, so the same subject naturally moves from near-certainty to genuine uncertainty across the battery.
- Some wrong answers are obviously dominated, while others are plausible competitors. That makes the *distribution* of beliefs over answers informative, not just the mode.
- The authors interpret Raven as a measure of fluid intelligence: solving novel problems rather than recalling learned facts.

Interpretation. Raven provides exactly the environment in which confidence should matter. A forced single answer throws away potentially important information about what the subject has ruled out and how uncertain the subject is.

2.2 Experimental samples and setting

- The main intelligence experiments were run in the ExCEN laboratory at Georgia State University in 2019 and 2022.
- Subjects were recruited by email with no information about the task or expected earnings and were randomly assigned across treatments.
- Each subject received a \$7 participation payment in addition to any task-specific earnings.
- All subjects had completed Raven Set I previously, either in an earlier linked experiment or earlier in the same session, so the basic format of the task was familiar.
- The “untimed” Raven treatments effectively allowed up to 90 minutes, and the average completion time was about 45 minutes.

The main fluid-intelligence treatment sample sizes are:

- Baseline (traditional): 55
- One Token Progressive: 95

- 80 Tokens Progressive: 82
- One Token Scrambled: 67
- 80 Tokens Scrambled: 61

The competitiveness extension uses within-subject modules with 43 subjects in the one-token arm and 45 subjects in the 80-token arm. The literacy extension uses 776 Georgia State undergraduates from Atlanta Federal Reserve survey waves in 2013, 2014, and 2016.

2.3 Belief elicitation and the QSR interface

Instead of forcing a single answer, the paper lets subjects allocate tokens across the eight possible Raven answers. The key scoring device is a quadratic scoring rule (QSR).

Let r_k denote the reported likelihood assigned to the interval or answer that turns out to be correct, and let $\{r_1, \dots, r_K\}$ be the full reported distribution. The canonical QSR score is

$$S = (2r_k) - \sum_{i=1}^K r_i^2.$$

The implemented payoff version is

$$a + b \left[(2r_k) - \sum_{i=1}^K r_i^2 \right],$$

where $a > 0$ and $b > 0$ scale earnings so that subjects never face losses.

Key features of the implementation:

- In the 80-token treatments, subjects allocate 80 tokens across the eight answers.
- If all 80 tokens are allocated to the correct answer, earnings are \$2 for that question.
- If 10 tokens are allocated to each answer, the subject guarantees \$1.13 regardless of the correct answer.
- The software shows the payoff consequences of reallocating tokens in real time, making the scoring logic transparent during the decision.

Interpretation. The QSR turns belief reporting into an incentivized choice problem. Full certainty can be rewarded, but caution can also be rational when the subject is genuinely unsure.

2.4 Main Raven treatments

The five principal Raven treatments differ along two dimensions: how much confidence can be expressed, and whether the question order provides a scaffold.

- **Baseline (traditional).** Pen-and-paper multiple choice with no QSR and \$5 for completion.

- **One Token Progressive.** Computerized, incentivized, and progressive in difficulty, but subjects can allocate only one token, so they must report the modal belief.
- **80 Tokens Progressive.** Same as One Token Progressive except subjects allocate 80 tokens and can report confidence.
- **One Token Scrambled.** Same one-token reporting, but question order is randomized rather than progressively easier-to-harder.
- **80 Tokens Scrambled.** Same as the 80-token progressive task, but with randomized question order.

Interpretation.

- Baseline versus One Token Progressive identifies the effect of salient incentives, keeping modal-belief reporting.
- One Token Progressive versus 80 Tokens Progressive identifies the effect of allowing confidence expression, holding computerization and reward levels fixed.
- Progressive versus Scrambled isolates the value of the ordering scaffold.

2.5 Outcome measures: accuracy and efficiency

At the subject level, the paper studies two core outcomes.

Accuracy.

- In the baseline and one-token treatments, accuracy is the fraction of questions answered correctly.
- In the 80-token treatments, accuracy generalizes to the fraction of tokens allocated to correct answers across the 36 questions.

Efficiency.

$$\text{Efficiency} = \frac{\text{realized earnings}}{\text{maximum possible earnings}}.$$

- The maximum possible earnings occur when all tokens are allocated to the correct answer on every question.
- Efficiency is therefore a welfare-relevant performance measure under the paper's incentivized environment.

The paper also classifies response types in the 80-token treatments:

- **Certain:** all 80 tokens on one answer;
- **Diffuse:** 10 tokens on each answer;
- **Unsure:** any nondegenerate allocation between those extremes.

2.6 Competition extension

The paper reuses scrambled Raven questions to revisit the competitiveness literature.

The design has six parts:

- Part 1: piece-rate compensation on 12 scrambled Raven questions over 10 minutes;
- Part 2: tournament compensation on a different 12 scrambled questions over 10 minutes;
- Part 3: choose whether a new 12-question round should be paid by piece rate or tournament;
- Part 4: choose whether *past* performance from part 1 should be paid by piece rate or tournament, removing the stress of performing under competition while preserving payoff risk;
- Part 5: elicit beliefs over piece-rate rank among four subjects;
- Part 6: elicit beliefs over tournament rank among four subjects.

The payoff structure is deliberately stark:

- piece rate pays up to \$2 per question;
- tournament pays \$8 per question only if the subject ranks first in the group.

Interpretation. Choosing to compete is a choice over risky lotteries, not just a taste-for-competition measure. That is why subjective probabilities and risk preferences must be measured.

2.7 Literacy extension

The literacy application adapts one of Lusardi and Mitchell’s “Big Three” inflation questions. Subjects answer an incentivized version of:

Imagine you have \$100 in a savings account, the interest rate is 1% per year, and inflation is 2% per year. After one year, how much purchasing power would you have on the initial \$100?

The correct purchasing-power answer is \$99.02. Subjects again report belief distributions across dollar bins using the QSR interface.

Interpretation. This lets the paper distinguish three things that ordinary financial-literacy surveys mix together:

- bias in the central answer,
- uncertainty around that answer,
- willingness to say “I do not know with enough confidence to bet on one option.”

3 Models and empirical specifications (all equations + interpretation)

3.1 Why forced modal answers mismeasure cognition

The paper’s organizing idea is simple: a single multiple-choice answer records only the *mode* of a subject’s belief distribution. But many cognitively relevant objects depend on more than the mode:

- whether the subject ruled out obviously bad options;
- whether the correct option received almost as much weight as the chosen answer;
- whether the subject is aware of being uncertain;
- whether the subject manages risky choices sensibly when answers have consequences.

Interpretation. A wrong but cautious answer is not the same as a wrong and confident answer. The standard test score treats them as identical.

3.2 QSR and the mapping from reports to beliefs

The descriptive interpretation of token allocations is that they reflect subjective beliefs. The paper is careful about this:

- If the subject is risk neutral, the QSR directly incentivizes truthful probability reports.
- Even under moderate expected-utility risk aversion, many observed reports coincide with true beliefs exactly: all-on-one, fully diffuse, or equal allocations over a subset.
- The cases where risk preferences can distort reports most are the intermediate nonuniform allocations.

Interpretation. The paper is not naively assuming that every token split is a perfect belief report. It argues instead that the assumption is exact in many empirically important cases and approximately correct in the rest.

3.3 Performance metrics and estimation

Because accuracy and efficiency are proportions, the paper estimates treatment effects with fractional regression models and reports average marginal effects.

The typical covariates include:

- gender,
- Black indicator,
- business-major indicator,
- atheist/no-religion indicator,
- age above 18,
- employment status.

Interpretation. The empirical design is experimental, but the paper still uses regression-based marginal effects to summarize how treatment effects interact with demographics.

3.4 The incentive effect versus the confidence effect

The first two treatment contrasts are conceptually different:

- **Baseline → One Token Progressive:** same modal-answer logic, but now with salient performance incentives.
- **One Token Progressive → 80 Tokens Progressive:** same computerized QSR environment and same \$2 maximum per question, but now the subject can express confidence.

This distinction explains the central result of section II:

- incentives improve *accuracy*;
- confidence expression improves *efficiency*.

Interpretation. If one keeps using hit-rate accuracy as the metric, the unconstrained choice problem can seem surprisingly unimpressive. The efficiency metric reveals why this is the wrong conclusion.

3.5 Progressive versus scrambled order as a scaffold test

The paper treats the progressive ordering of Raven questions as a cognitive scaffold. If questions arrive from easy to hard, the subject gains information from the ordering itself.

The scrambled treatments therefore ask whether the female and Black gains under 80-token reporting simply reflect exploiting that scaffold rather than better calibrated confidence reporting.

Interpretation. If the main gender result vanished under random ordering, then the paper would only be showing that women exploit the test designer’s ordering clues better. The result does *not* vanish.

3.6 Welfare cost of forcing the modal belief

A major conceptual step is to evaluate how costly it is to force a subject to behave as if her modal belief were her entire belief.

The welfare calculation uses:

- recovered beliefs from the 80-token reports,
- the implied lottery over QSR earnings when the subject is forced to one-token reporting,
- estimated risk preferences.

The paper reports certainty equivalents (CEs) under both expected utility theory (EUT) and rank-dependent utility (RDU). Under EUT, utility is constant-relative-risk-aversion:

$$u(x) = \frac{x^{1-r}}{1-r}.$$

Under RDU, probabilities are transformed using a Prelec weighting function:

$$q(p) = \exp\{-h(-\ln p)^J\}.$$

Interpretation. The paper is asking a normatively sharper question than “does the score change?” It asks how much welfare the subject loses when the experimenter suppresses confidence information.

3.7 Competition as lottery choice

The standard competitiveness literature interprets the choice of tournament pay as willingness to compete. This paper reframes that choice as a comparison of two risky lotteries.

The key welfare objects are:

- the payoff from piece-rate compensation;
- the payoff from winning the tournament;
- the subjective probability of winning the tournament;
- the subject's risk preferences.

The certainty equivalent of each compensation scheme solves

$$u(CE) = EU \quad \text{or} \quad u(CE) = RDU.$$

Interpretation. Once the tournament is recognized as a lottery with a high chance of zero payoff, choosing the safe piece-rate contract can be the welfare-maximizing choice. This is why the paper argues that the “women avoid competition” conclusion is based on the wrong metric.

3.8 Literacy as bias *and* confidence

In the literacy application, the paper argues that absolute bias alone is insufficient. A subject can be wrong in two very different ways:

- **wrong and diffuse:** poor point estimate but openly admitted uncertainty;
- **wrong and confident:** poor point estimate with complete conviction.

Interpretation. The second type is normatively more dangerous. It is the subject who may “bet the house” on a wrong belief and therefore most needs corrective information or scaffolding.

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the main effects

The paper does *not* use an instrument in the usual IV sense. The identifying variation comes from randomized assignment across experimental treatments that differ in:

- whether incentives are salient,
- whether confidence can be expressed,
- whether the progressive ordering scaffold is available.

Interpretation. This is a clean experimental paper. Causal identification comes from treatment assignment, not from quasi-experimental timing or instruments.

4.2 Why the incentive comparison is reasonably credible

The baseline-to-one-token comparison changes both incentives and the delivery mode (pen-and-paper versus computerized QSR). The paper openly acknowledges that this comparison is cleanest for incentives only if computerization does not itself alter performance.

The authors defend this assumption by:

- identical recruitment and random assignment across treatments;
- evidence from related studies that paper and computer Raven performance are similar for university-level subjects.

Interpretation. The baseline comparison is persuasive but not perfect. The more internally clean comparison is one-token versus 80 tokens within the computerized environment.

4.3 How the paper isolates confidence from everything else

One Token Progressive and 80 Tokens Progressive are identical in almost every relevant respect:

- same subject pool,
- same software,
- same progressive Raven sequence,
- same \$2 maximum per question,
- same time environment.

The difference is purely whether subjects can report only a modal belief or a full confidence distribution.

Interpretation. This is the paper's cleanest design contrast for the value of confidence measurement.

4.4 How the paper isolates the scaffold effect

The progressive and scrambled comparisons keep the scoring rule and incentives fixed while changing only the question order.

Interpretation. This allows the paper to separate two ideas:

- confidence reporting helps because it is a better measurement technology;
- confidence reporting helps because it is bundled with an ordering scaffold.

The persistence of the female and Black gains in the scrambled task means the main result is not purely an artifact of the progressive order.

4.5 Why the welfare calculations are unusually credible

The competition and modal-belief welfare analyses require more than simple treatment means. The paper separately measures:

- subjective beliefs,
- risk preferences,
- realized payoffs,
- and chosen actions.

This allows the authors to evaluate whether the *chosen* action improved or reduced welfare for the person choosing it.

Interpretation. Many papers on competition or confidence stop at observed choice frequencies. This paper goes one step further and asks whether the observed choices were normatively sensible.

4.6 What the paper can and cannot distinguish

The paper is strong on the claim that confidence reporting changes measured intelligence, competitiveness, and literacy. It is weaker on two finer distinctions:

- It cannot fully separate “actually being less certain” from “being more willing to admit uncertainty.”
- It cannot fully separate the interface as a measurement device from the interface as a productivity-enhancing scaffold.

The second point is deliberate. The authors explicitly argue that if the interface makes people think more effectively, that is itself part of the relevant economic object.

4.7 External validity and sample limits

The main Raven and competitiveness samples are Georgia State undergraduates, not nationally representative adults. The literacy extension is also based on Georgia State students.

Interpretation. The paper’s claims are strongest as a methodological critique of how traits are measured, not as a claim about the exact population-level magnitude of gender differences in the United States.

4.8 Relation to the literature

The paper intervenes in three literatures at once:

- intelligence testing, especially Raven-based measures of fluid intelligence;
- experimental “willingness to compete” studies;
- financial-literacy surveys using forced multiple-choice responses and “do not know” options.

Its recurring argument is that each literature relies too heavily on observable point responses and too little on incentivized confidence reporting and welfare evaluation.

5 Tables and figures

5.1 Figure 2: Raven scores with no salient rewards

Read:

- The figure plots the fraction correct across the 36 Raven questions for the Georgia State sample without salient performance rewards, alongside comparison samples from Arthur and Day (1994) and Gignac (2018).
- The broad downward trend confirms the progressive increase in difficulty.
- The Georgia State sample performs somewhat worse than the comparison samples, especially through the middle of the battery.

Economic message. The Raven battery naturally moves subjects from confident knowledge to uncertainty. That is exactly why a confidence-sensitive measurement approach matters.

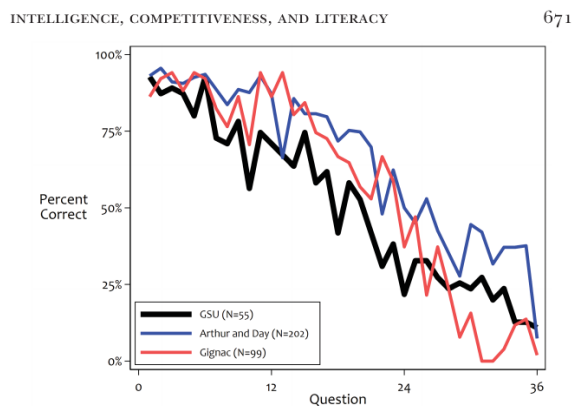


FIG. 2.—Raven scores with no salient rewards.

Figure 1: Figure 2 from the paper: Raven scores with no salient rewards.

5.2 Figure 4 and Table 1: the QSR interface and treatment architecture

Read:

- Figure 4 shows two key displays of the 80-token interface: a nondegenerate confidence report and the fully diffuse 10/10/10/10/10/10/10/10 allocation that guarantees \$1.13.
- Table 1 lays out the full experimental design. Panel A contains the five core Raven treatments. Panel B adds the six within-subject competitiveness treatments.

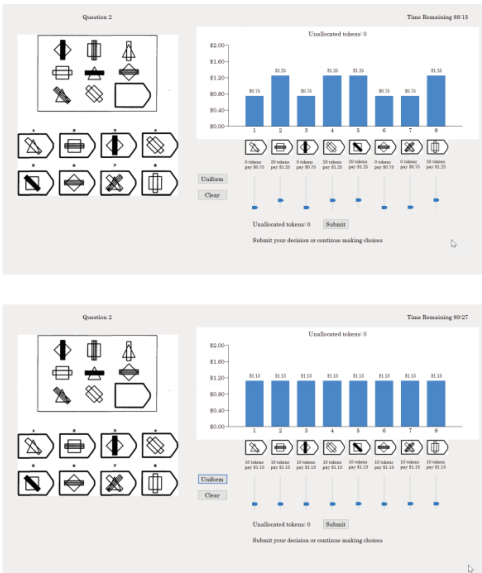


FIG. 4.—Belief elicitation interface with two possible displays of confidence.

there are no reallocations to be made. An exception occurs if the time limit is reached, in which case the individual knows that she would re-

- The table makes clear that the central comparisons are built from deliberately paired designs: one token versus 80 tokens, and progressive versus scrambled.

Economic message. The paper’s identification is visible in the treatment table itself. It is designed to decompose incentives, confidence expression, scaffolding, and tournament risk.

5.3 Figures 5 and 6: incentives improve accuracy, confidence improves efficiency

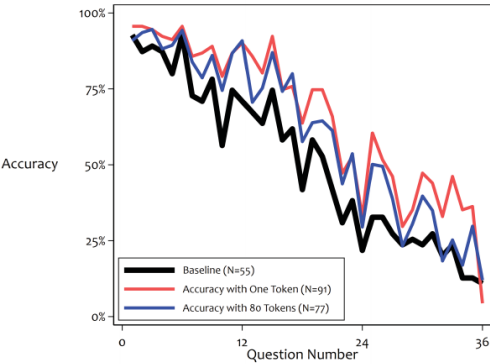


FIG. 5.—Average accuracy and incentives with progressive Raven problems. Accuracy is measured by the percent of tokens allocated to the correct answer. Beliefs are assumed to be the same as elicited reports.

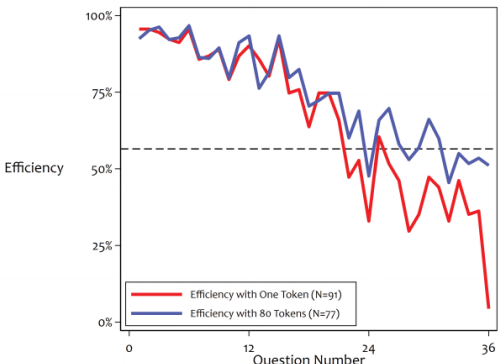


FIG. 6.—Average efficiency with progressive Raven problems. Efficiency is measured by the percent of realized earnings compared with maximum possible earnings. A uniform token allocation ensures earnings of \$1.13 and efficiency of 56.5%. Beliefs are assumed to be the same as elicited reports.

Read:

- Figure 5 shows that adding salient incentives raises average Raven accuracy substantially relative to the traditional baseline.
- The one-token and 80-token progressive treatments have broadly similar accuracy paths, so confidence reporting alone does not dramatically raise raw hit rates.
- Figure 6 shows the real payoff gain from confidence reporting: the 80-token treatment stays well above the one-token treatment on efficiency, especially in the hard last third of the battery.

Economic message. The central methodological point of section II is visible here. If the researcher insists on accuracy alone, the gain from confidence reporting looks small. If the researcher uses the right metric—earnings efficiency under the incentivized task—the gain is large.

5.4 Figure 7: why the gender conclusion flips

Read:

- Panel A shows that, under the traditional baseline, women look worse on accuracy.
- Panel B shows that moving from baseline to one-token incentives improves women’s accuracy, but not enough to overturn the conventional conclusion.
- Panel C is the decisive panel: moving from one-token to 80-token reporting produces a clear positive efficiency effect for women, and a positive effect for Black subjects as well.

Economic message. This is the figure that justifies the paper’s headline claim. The apparent female disadvantage in intelligence is not robust to measuring performance with incentivized confidence rather than forced modal answers.

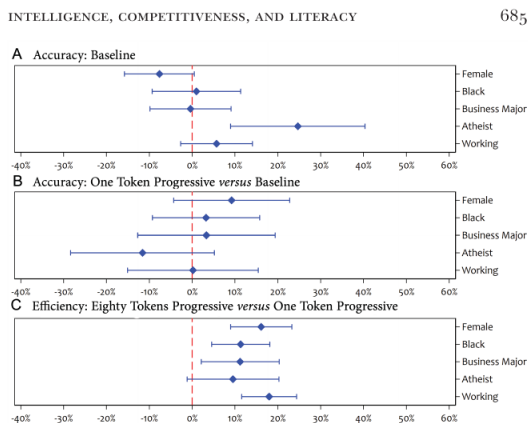


FIG. 7.—Effects of demographics with progressive Raven problems. All results used the original progressive order of the Raven problems. Average marginal effects are from fractional regression.

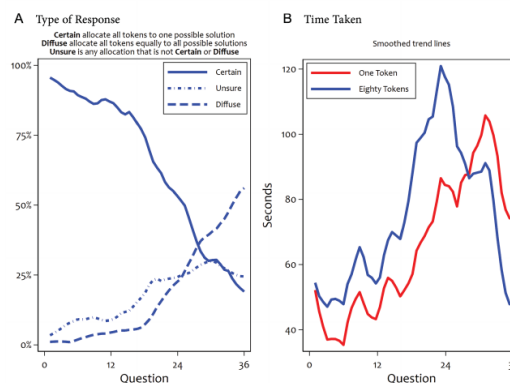


FIG. 10.—Time trends of time for response and for type of response.

5.5 Figure 10: what confidence reporting changes inside the task

Read:

- The share of *certain* allocations falls steadily from roughly 95% early in the battery to about 20% by the end.
- *Unsure* and especially *diffuse* allocations rise as the questions become harder; diffuse responses dominate the last questions.
- Time use also changes: one-token subjects keep spending time deep into the hard questions, whereas 80-token subjects begin cutting time earlier because they can rationally shift toward diffuse allocations.

Economic message. Confidence reporting changes behavior inside the cognitive task itself. Subjects do not just reveal uncertainty; they also allocate effort differently once they are allowed to express uncertainty.

5.6 Figures 11–13: scrambled Raven and the welfare cost of suppressing confidence

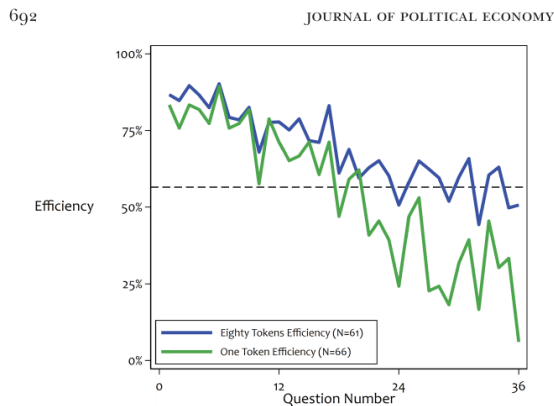


FIG. 11.—Average efficiency with scrambled Raven problems. Efficiency measured by the percent of realized earnings compared with maximum possible earnings. A uniform token allocation ensures earnings of \$1.13 and efficiency of 56.5%. Question number is from the original progressive order. Beliefs are assumed to be the same as elicited reports.

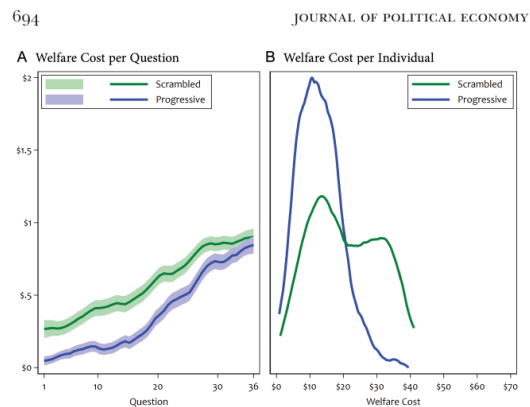


FIG. 13.—Expected welfare cost to respondent of being required to report only modal belief, evaluated using RDU risk preferences for each subject and using recovered beliefs from the 80 tokens task.

Read:

- Figure 11 shows that the efficiency dominance of 80-token reporting survives when Raven questions are scrambled.
- Figure 12 shows that the positive female and Black effects remain in the scrambled treatment, so the result is not just exploitation of the progressive ordering scaffold.
- Figure 13 quantifies the welfare loss from forcing modal-belief reports. The loss rises with question difficulty and is materially larger in the scrambled condition.
- Average welfare losses per person are about \$13.48 under progressive ordering and \$20.88 under scrambled ordering, a difference of roughly \$7.40.

Economic message. Confidence reporting is not just statistically useful. It has a concrete certainty-equivalent value for subjects, and that value is larger when the test designer removes the ordering scaffold.

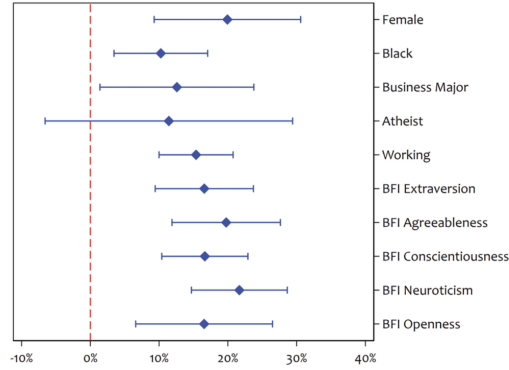


FIG. 12.—Effects of 80 tokens on efficiency with scrambled Raven problems. Average marginal effects from fractional regression, pooling 80 token and one token results, with a dummy variable for the 80 token condition and solely comparing results within the scrambled treatment.

5.7 Table 2 and Figures 14–16: competitiveness measured by welfare

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TABLE 2
WELFARE EVALUATION OF WILLINGNESS-TO-COMPETE DECISIONS

	Risk Neutral		EUT		RDU	
	Male	Female	Male	Female	Male	Female
A. Payoffs for 80 Tokens						
Piece rate (\$)	1.37	1.36	1.37	1.36	1.37	1.36
Tournament win (\$)	5.66	5.72	5.66	5.72	5.66	5.72
Tournament loss (\$)	0	0	0	0	0	0
B. Probabilities for 80 Tokens						
Piece rate	1	1	1	1	1	1
Tournament win	.26	.28	.26	.28	.26	.28
Tournament loss	.74	.72	.74	.72	.74	.72
C. Risk-Preference Parameters						
Utility parameter r	0	0	.6	.56	.71	.7
PWF parameter η	1	1	1	1	.97	.9
PWF parameter ϕ	1	1	1	1	1	.92
D. Certainty Equivalents for 80 Tokens						
Piece rate (\$)	1.37	1.36	1.37	1.36	1.37	1.36
Tournament (\$)	1.47	1.60	.87	.98	.09	.18
E. Expected Welfare Gain for 80 Tokens						
Piece-rate compensation (\$)	−.10	−.24	+.50	+.38	+.128	+.118
Tournament compensation (\$)	+.10	+.24	−.50	−.38	−.128	−.118
F. Expected Welfare Gains for One Token						
Piece-rate compensation (\$)	+.11	+.20	+.51	+.62	+.99	+.109
Tournament compensation (\$)	−.11	−.20	−.51	−.62	−.99	−.109

NOTE.—Parameter values for payoffs, subjective probabilities, and risk preferences are documented in app. C.

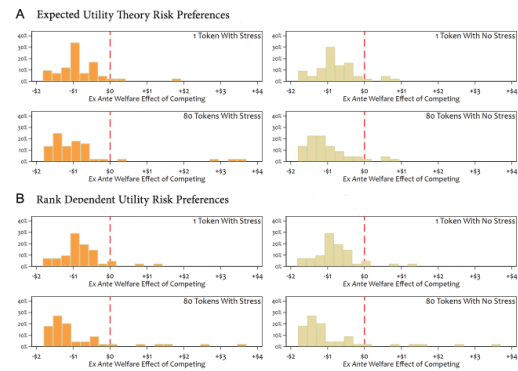


FIG. 14.—Ex ante welfare effects of the decision to compete.

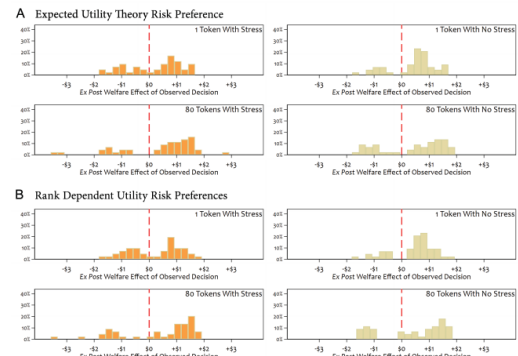


FIG. 15.—Ex post welfare effects of the decision to compete (or not).

Read:

- Table 2 shows that under risk neutrality the tournament may slightly dominate the piece rate, but once EUT or RDU risk aversion is used, the piece-rate contract has the higher certainty

equivalent for both men and women.

- In the one-token case, even risk-neutral subjects should choose the piece rate.
- Figure 14 shows that for the vast majority of subjects, the ex ante welfare effect of choosing to compete is negative.
- Figure 15 shows many ex post welfare losses from observed competition choices, because many subjects compete when their own beliefs and risk preferences imply that they should not.
- Figure 16 shows that women are only modestly optimistic in the 80-token case, while men are closer to the diffuse-prior benchmark on average.

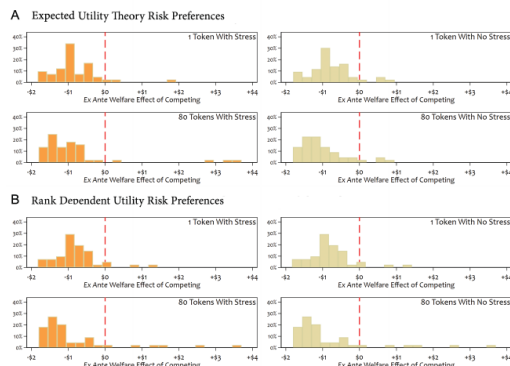


FIG. 14.—Ex ante welfare effects of the decision to compete.

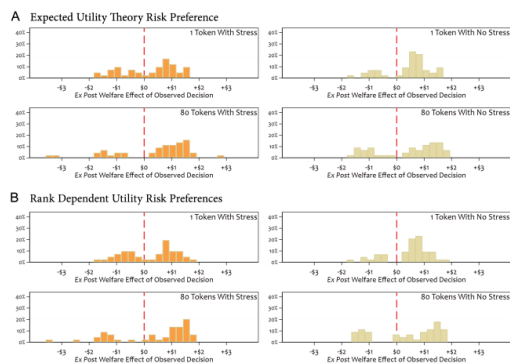


FIG. 15.—Ex post welfare effects of the decision to compete (or not).

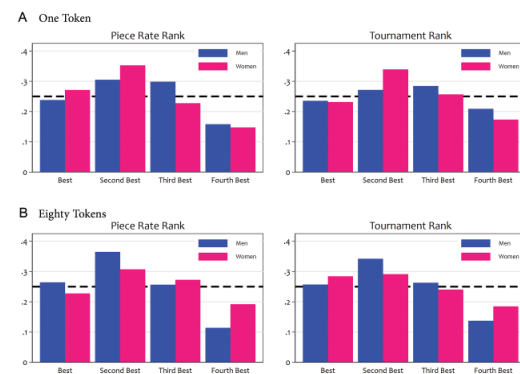


FIG. 16.—Recovered beliefs on the probability of personal performance ranks by men and women.

Economic message. This section overturns the usual reading of tournament-choice experiments. The interesting puzzle is not that women avoid competition. It is that many men select a dominated risky contract once their own subjective risk perceptions are taken seriously.

5.8 Figures 17 and 18: literacy is about confidence calibration, not just point bias

Read:

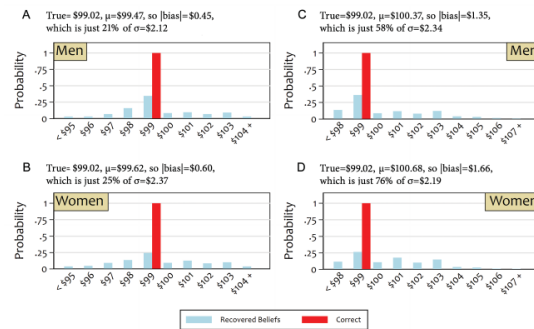


FIG. 17.—Bias and confidence in the literacy of men and women on the inflation question.

the diffuse belief suggests to us that this is a subject who is likely to start searching for a scaffold to make a better decision. Whether that scaffold is the advice of the financial planner, the internet, or a partner, the evidence of the diffuse belief should only be taken as evidence of “naked, Robinson Crusoe without bars on his phone” illiteracy.

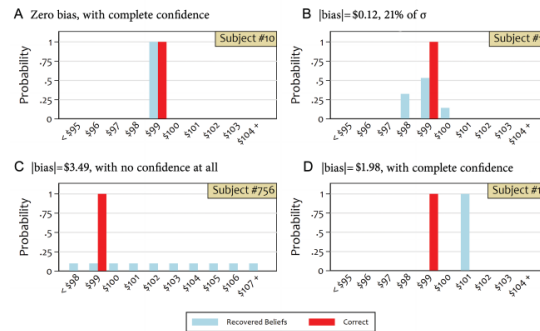


FIG. 18.—Literacy and confidence: the good, the bad, and the ugly.

- Figure 17 shows that both men and women have small point bias on the inflation question, with women’s mean response slightly farther from the correct \$99.02 than men’s.
- But those biases are small relative to the spread of reported beliefs, so the paper interprets them as economically minor once confidence is respected.
- Figure 18 then distinguishes four types of respondent: correct-and-confident, near-correct-but-cautious, badly biased but fully diffuse, and badly biased with complete confidence.
- The most worrying type is the confidently wrong respondent, not the diffuse respondent who effectively admits “I do not know.”

Economic message. This is the paper’s cleanest challenge to the financial-literacy literature. A greater willingness to avoid a falsely precise answer should not automatically be coded as lower literacy.

6 Short summary

- The paper argues that conventional intelligence tests mismeasure cognition because they ignore confidence and usually ignore salient incentives.
- In Raven experiments, salient incentives raise accuracy, but allowing confidence reports mainly raises earnings efficiency rather than raw hit rates.
- Standard scoring makes women look worse on Raven accuracy; confidence-sensitive scoring overturns that inference and gives women an efficiency advantage.
- The same qualitative result survives when the progressive order of Raven questions is scrambled, so the conclusion is not driven only by the original test ordering.
- When willingness-to-compete experiments are evaluated using subjective probabilities and risk preferences, women do not appear insufficiently competitive; many men appear excessively willing to choose risky tournaments.
- In the literacy application, women’s apparently lower literacy is reinterpreted as more appropriate expression of uncertainty rather than simply larger ignorance.
- The paper’s methodological claim is broad: use incentivized belief distributions and welfare-relevant metrics when measuring cognition.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that measuring intelligence by forced right-or-wrong answers is incomplete and often misleading. Salient incentives matter, and so does the ability to report confidence. Once those features are built into the measurement environment, the apparent gender ranking of cognitive performance changes sharply: women perform better than standard modal-belief scoring suggests, and in efficiency terms they can outperform men. The same logic carries over to celebrated claims about competition and financial literacy.

7.2 Economic intuition

A knowledge claim is usually a risky choice, not a certainty. When someone chooses among multiple possible answers, the economically relevant object is their full belief distribution and how they manage the risk attached to it. A person who is slightly wrong but openly uncertain is very different from a person who is equally wrong and fully confident. The paper's elicitation interface lets subjects price that uncertainty, respond to it, and sometimes even think better because the interface itself supplies a scaffold.

7.3 Takeaway

The takeaway is not that women simply need “more confidence,” nor that intelligence is a fixed inner trait that tests merely reveal. The paper argues that *appropriate confidence* is part of cognition itself, and that measurement environments shape what intelligence, competitiveness, and literacy look like. For empirical work, that means hit rates and forced choices are often too crude. For policy, it means that helping people recognize and communicate uncertainty can improve both measured performance and welfare.